

Problem — thin-film encapsulation (TFE) of perovskite PV:
 geometry design space (d_{org} , d_{inorg} , n_{pairs}) vs six conflicting objectives
 (T80 lifetime · device temperature · transmittance · cost · weight · thermo-mech durability)

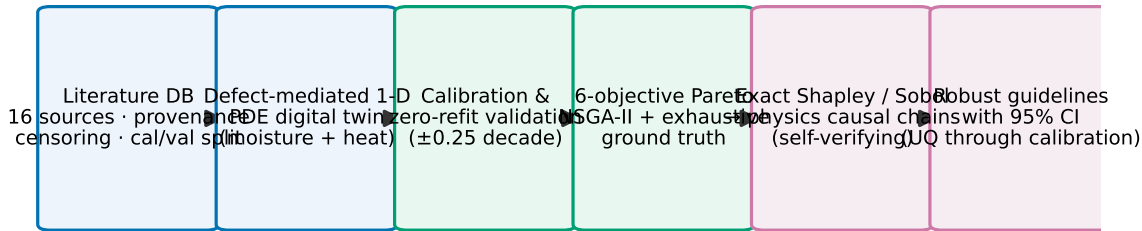


Fig. 1 | Physics-informed digital-twin framework: every arrow is a validated, number-carrying step (Steps 1–8 of this work).